PRECAUTION [ANTILOCK BRAKE SYSTEM (ABS)]

id040308804100

1. The ABS warning light and/or brake system warning light illuminate even when the system is normal.

Warning lights that may illuminate	Cases in which the light may illuminate	Conditions in which the light will go out	ABS and EBD control
• ABS warning light • Brake system warning light	Only the front wheels rotate under the following condition while jacked up, stuck, or on the chassis dynamometer: • Detected **8 times** for a continuous **20 s** while at a vehicle speed of **20 km/h {12 mph} or more**. (One detection period is when ignition is switched from ON to off.)	After turning off the ignition once and then restarting the engine, the vehicle is driven at **10km/h {6.2 mph} or more**, and then normal operation is verified.	• ABS: Cuts control. • EBD: Cuts control.
	Parking brake is not fully released while driving.		
	Brake drag.		
	Sudden acceleration/ deceleration.		
	Left/right or front/rear tires are different. (Size, radius, tire pressure, or wear is other than that listed on tire label.)		
	Battery voltage at ABS HU/CM terminal of voltage source drops **below about 9.5 V**.	Battery voltage rises **above about 9.5 V**.	• ABS: Operates control. • EBD: Operates control.
Brake system warning light	Brake fluid amount is low.	Brake fluid level is within the specification.	• ABS: Operates control. • EBD: Operates control.

2. Precautions during servicing of ABS. The ABS is composed of electrical and mechanical parts. It is necessary to categorize malfunctions as being either electrical or hydraulic when performing troubleshooting.
   (1) Malfunction in electrical system
      - The control module has an on-board diagnostic function. With this function, the ABS warning light and/ or brake system warning light will come on when there is a problem in the electrical system.
        Also, past and present malfunctions are recorded in the control module. This function can find malfunction that do not occur during periodic inspections. Connect the M-MDS (IDS) to the DLC-2. The stored malfunctions will be displayed in the order of occurrence. To find out the causes of ABS malfunction, use these on-board diagnostic results.
      - If a malfunction occurred in the past but is now normal, the cause is likely a temporary poor connection of the harness. The control module usually operates normally. Be careful when searching for the cause of malfunction.
      - After repair, it is necessary to clear the DTC from the control module memory. Also, if the ABS related parts have been replaced, verify that the no DTC has been displayed after repairs.
      - After repairing the ABS wheel-speed sensor or ABS sensor rotor, or after replacing the control module, the ABS warning light may not go off even when the ignition is switched ON. In this case, drive the vehicle at a speed of **more than 10 km/h {6.2 mph}**, make sure the ABS warning light goes off, and then clear the DTC.
      - When repairing, if the ABS related connectors are disconnected and the ignition is switched ON, the control module will mistakenly detect a fault and record it as a malfunction.
      - To protect the control module, make sure the ignition is switched off before connecting or disconnecting the control module connector.
   (2) Malfunctions in hydraulic system
      - Symptoms of a hydraulic system malfunction are similar to those of a conventional brake malfunction. However, it is necessary to determine if the malfunction is in a ABS component or the conventional brake system.
      - The hydraulic unit contains delicate mechanical parts. If foreign matter gets into the component, the ABS may fail to operate. Also, it will likely become extremely difficult to find the location of the malfunction in the event that the brakes operate but the ABS does not. Make sure foreign matter does not penetrate the hydraulic unit when servicing the ABS (e.g. brake fluid replacement, pipe removal).

Intermittent Concern Troubleshooting
Vibration method

- If a malfunction occurs or worsens while driving on a rough road or the engine vibrates, perform the following steps.

Note

- There are several reasons why the vehicle or engine vibration could cause an electrical malfunction. Inspect the vehicles for the following:
  - — Connectors not fully seated.
  - — Wiring harnesses not having full play.
  - — Wires laying across brackets or moving parts.
  - — Wires routed too close to hot parts.
- An improperly routed, improperly clamped, or loose wiring harness can cause it to become pinched between parts.
- The connector joints, points of vibration, and places where wire harnesses pass through the firewall, and the body panels are the major areas to be checked.

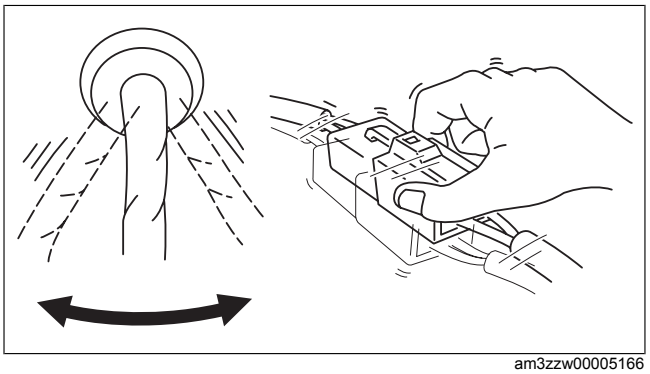
Inspection method for switches, sensor connectors, and wires

1. Connect the M-MDS to the DLC-2.
2. Switch the ignition ON (engine off).

Note

- If the engine starts and runs, perform the following steps while idling.

3. Access PIDs for the switch or sensor being inspected.
4. Turn the switch on manually.
5. Slightly shake each connector or wiring harness vertically and horizontally while monitoring the PID.
   - If the PID value is unstable, inspect for poor connection.



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Inspection method for switches and sensors

1. Connect the M-MDS to the DLC-2.
2. Switch the ignition ON (engine off).

Note

- If the engine starts and runs, perform the following steps while idling.

3. Access PIDs for the switch or sensor being inspected.
4. Vibrate the sensor slightly with a finger.
   - If PID value is unstable or a malfunction occurs, check for a poor connection or poorly mounted sensor.

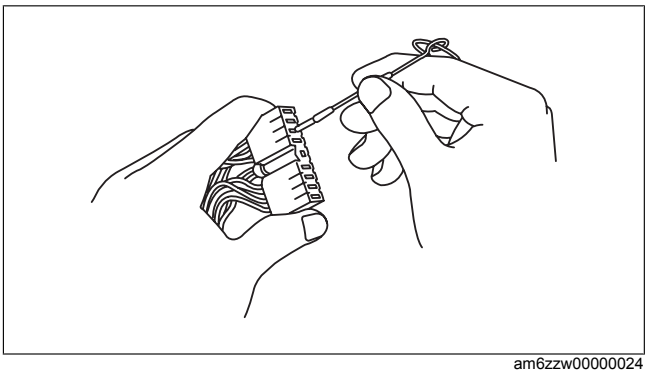
Malfunction data monitor method

1. Perform malfunction reappearance test according to malfunction reappearance mode and malfunction data monitor. The malfunction cause is found in the malfunction data.

Connector terminal check method

1. Check the connection condition of each female terminal.

2. Insert the male terminal; fit male terminal to female terminal and verify if malfunction is in female terminal.



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